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Life-cycle greenhouse gas emissions and the associated carbon-peak strategies for PS, PVC, and ABS plastics in China

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ABSTRACT

The plastic industry is a high carbon emission industry in China. Previous studies mainly focused on the flows and stocks of polystyrene (PS), polyvinyl chloride (PVC), and acrylonitrile-butadiene-styrene (ABS) plastics. However, there is limited detailed information on the GHG emissions of PS, PVC, and ABS in China and the key paths of carbon peak in 2030. Therefore, we established the GHG accounting model of PS, PVC, and ABS based on dynamic material flow analysis and life cycle assessment, and further analyzed the carbon peak path and strategy in 2030. The results show that the main source of GHG is the production and manufacturing stages. Notably, the key paths of GHG emissions are the manufacturing process of PS STP&C and foam plastics, PVC film and STP&C, and ABS STP&C. Moreover, the cumulative GHG emissions of these plastics reached 230.34, 677.43, 143.12 Tg CO_2e in 2007-2017, respectively. From 2007 to 2017, the trends of GHG emissions from the PS, PVC, and ABS had their characteristics. PVC had the largest GHG emissions. Different from PVC and ABS, GHG emissions from PS showed a trend of slow rise and then slow decline in this period. According to the current trend, the life-cycle GHG emissions of PS, PVC, and ABS would not reach the carbon peak in 2030. Under some mitigation strategies, the carbon peak would be achieved before 2030. This study clarifies the main sources of life-cycle GHG emissions of these plastics, and reveals the path and strategy of carbon peak in 2030.

1. Introduction

Greenhouse gas (GHG) emissions from human activities into the atmosphere are considered to be the main cause of global warming (Solomon et al., 2007). In 2016, 175 countries signed the Paris Agreement, the main purpose of which is to control the global average temperature rise within 2°C and strive to achieve zero GHG emissions in the second half of the 21st century. China is one of the world's largest carbon-emitting countries. In 2017, total carbon emission in China accounted for about 28.27% of the global total and had not yet reached its peak (International Energy Agency, 2019). In 2020, China proposed to strive to achieve the goal of carbon peak by 2030 and carbon neutrality by 2060. China plans that the energy efficiency of key industries would reach the international advanced level by 2030, and the energy consumption per unit of GDP would be greatly reduced; carbon emissions per unit of GDP decrease by more than 65% compared with 2005; the consumption proportion of non-fossil energy reaches about 25%; and the carbon emissions reach the peak and decreased steadily (The State Council, 2016, 2021). The proportion of non-fossil energy consumption would reach more than 80%, and the goal of carbon neutrality would be successfully achieved by 2060 (The State Council, 2016, 2021). The plastic industry is an industry with high energy consumption and carbon emissions. Plastic products are expected to contribute 15% to global GHG emissions by 2050 (Ellen MacArchur Foundation, 2016). GHG is emitted in the plastics life cycle (including the production (P), processing and manufacturing (PM), recycling, landfill, and incineration) (Hamilton et al., 2019). Plastics can be widely used in various industries (such as packaging and transportation). In

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recent years, the yield of plastics is amazing, and 8.3 billion tonnes of plastics were produced worldwide (Geyer et al., 2017). The huge production of plastics has brought a lot of GHG emissions. In the United States, the emissions from the exploitation and transportation of fossil fuels (mainly shale gas) related to plastics production were 9.5-10.5 million tonnes of carbon dioxide equivalent (CO_2e) in 2015 (Hamilton et al., 2019). The research showed that plastic consumption in the United States contributed to the GHG emissions of 104 million metric tons CO_2e (Nicholson et al., 2021). By 2030, the net GHG emissions from the incineration of plastic packaging waste would increase to 84 million tonnes (Ellen MacArthur Foundation, 2016b; Walker and McKay, 2021). The processes of production and incineration for plastics contributed more than 850 million tonnes of GHG, equivalent to the GHG emissions of 189 coal-fired power plants (500 megawatts) in 2019 (Hamilton et al., 2019).

At present, GHG emissions associated with plastics have attracted extensive attention. For example, Dormer et al. (2013) carried out cradle-to-grave research on plastic trays, deeply analyzed the carbon footprint of their products.(Zheng and Suh, 2019) analyzed global GHG emissions from both traditional plastics and bioplastics throughout their life cycle. They found that the global GHG emissions of traditional plastics are projected to grow to 650 million tonnes of CO_2e by 2050. Gomes et al. (2019) carried out a systematic review on the LCA studies of PET packaging materials.Brizga et al. (2020) analyzed the potential environmental consequences of replacing petrochemical packaging plastics with bioplastics (in terms of GHG emissions, land and water footprints) based on a review of LCA studies and calculations to assess the footprint of such alternatives.Su et al. (2020) quantified the environmental impact (CO_2e) of raw material production of express packaging, box manufacturing, and waste management. It showed that there was a great opportunity to reduce the consumption and impact of packaging materials.Meys et al. (2021) constructed a bottom-up model, which studied the pathways to net-zero GHG emissions of 90% global plastics. The results showed that net-zero GHG emissions could be achieved by combining the efficient utilization of carbon dioxide (CO_2) and biomass.(Alsabri and Al-Ghamdi, 2020) analyzed the LCA and carbon footprint of the production process of polyvinyl chloride (PVC), polyethylene (PE), and polypropylene (PP) plastics to find the environmental impact leading to global warming. It is found that PVC had higher energy consumption and carbon emissions than other plastics. In conclusion, although these studies involve the carbon footprint of plastics, however, there are few studies on life-cycle GHG emissions from the most commonly used polystyrene (PS), PVC, and acrylonitrile-butadiene-styrene (ABS) plastics in China, especially on the carbon peak path and strategy of these plastics in 2030. In addition, most of the previous studies on plastic carbon footprint are based on LCA, which can assess the level of environmental impacts throughout its life cycle (from material generation to waste treatment), which cannot be revealed by material flow analysis (MFA).(García-Guaita et al., 2018) However, the detailed mass flow and balance in the MFA system can provide inventory or activity level for LCA (Allesch and Brunner, 2015; Wang et al., 2022), which will solve the difficulty of obtaining the activity level required for GHG emissions accounting in the past. Therefore, this study will explore the GHG emissions of the plastics life cycle in combination with LCA aCPPIA, 2018nd MFA.

China is the world's largest producer and consumer of plastics (Chen et al., 2019; PlasticsEurope, 2018). PS, PVC, and ABS are the most widely used plastics in China, and the production and consumption of these three plastics reached 24.70 and 26.77 million tonnes respectively in 2017, which accounted for more than half of that of commonly used plastics (Chen et al., 2019;CPCIA, 2018;). However, the detailed information on the life-cycle GHG emissions of PS, PVC, and ABS plastics, especially the GHG emissions in the production (P), processing and manufacturing (PM), waste management and recycling (WM&R) stages in China, as well as the key path of carbon peak in 2030, is limited. This leads to the fact that the key path of emission reduction for PS, PVC, and ABS plastics in the life cycle under the background of carbon neutralization is unknown, and effective strategies of emission reduction cannot be proposed. To solve this problem, this study combines dynamic MFA with LCA to establish the GHG emissions accounting model of PS, PVC, and ABS in China. According to the model, the GHG emission characteristics of PS, PVC, and ABS plastics from 2007 to 2017 are analyzed, and the carbon peak in 2030 is further explored. This study mainly has two innovations: 1) It reveals the characteristics of GHG emissions from the PS, PVC, and ABS life cycle from the time series; 2) According to the historical emission trend, it further explores the key path and strategy for realizing the carbon peak of PS, PVC, and ABS plastics. This study would clarify the main sources of GHG emissions in the PS, PVC, and ABS plastics life cycle, and put forward policies to improve the GHG emissions from PS, PVC, and ABS plastics, which would provide scientific support for the PS, PVC, and ABS plastics industry to achieve the goal of carbon peak and neutralization.

2. Methodology

2.1. Spatial and temporal scale and extent

The system boundary of this study includes the time and space boundaries. The temporal scale is 2007-2017. The spatial scale is the Chinese mainland (excluding Hong Kong, Macao, and Taiwan). The material flows and GHG accounting system for PS, PVC, and ABS plastics constructed in this study mainly include four stages: P, PM, use, WM&R stages. Our previous work has established the primary material flows model of plastics (Chu et al., 2021a). Based on this model, some revisions are made to make it more suitable for analyzing the GHG emissions of PS, PVC, and ABS plastics in the whole life cycle. See Fig. S1 and supplementary materials (SI, including unit process, system boundary, life cycle inventory, etc.) for details. In this study, GHG emissions from three life cycle stages of PS, PVC, and ABS plastics were considered: first, P stage, including all activities from the cradle of PS, PVC, and ABS to the gate of resin production plant (production, general purpose, high impact, granulate, and bulk polymerization); second, PM stage, covering the manufacturing process of transforming resin into final plastics products (injection molding, blow molding, and extrusion conversion technologies); and finally, WM&R stage, referring to the incineration, landfill, discarded/untreated and recycling process of PS, PVC, and ABS plastic wastes. Due to the unavailability of data, it was not considered the use stage in this study (Fig. 1).

2.2. Life-cycle GHG emissions of fossil PS, PVC, and ABS plastics

This study combined LCA and MFA methods to analyze the GHG emissions of PS, PVC, and ABS in the whole life cycle. Based on the law of mass balance, an MFA can effectively track the flows of materials in the system and has been widely used in the flows of elements or plastics (Brunner and Rechberger, 2016; Chu et al., 2019; Chu et al., 2021b; Graedel, 2019; Van eygen et al., 2018). When analyzing the flows of PS, PVC, and ABS life cycle, each stage in the system meets the mass balance. We used dynamic MFA to quantify the annual production and the flow and waste volume of PS, PVC, and ABS plastics, which can provide activity level data of each process (technology), to prepare for the accounting of total GHG emissions (Wang et al., 2021). Detailed mass flow and balance in an MFA system can provide an informed inventory or activity level for LCA (Allesch and Brunner, 2015; Wang et al., 2022). Therefore, the material or waste flows calculated by MFA are used as the activity level required to calculate GHG in this study (i.e., the value of Q in formula 3). An LCA is widely used to analyze the environmental impact and GHG emissions of products in the life cycle (Chu et al., 2022, Posen et al., 2017; Wu et al., 2019). According to the provisions of IPCC and other studies, we adopted a process-based approach to quantify the GHG emissions of PS, PVC, and ABS plastics during their life cycle (assuming the sum of the research processes) (IPCC, 2006; Milford et al.,

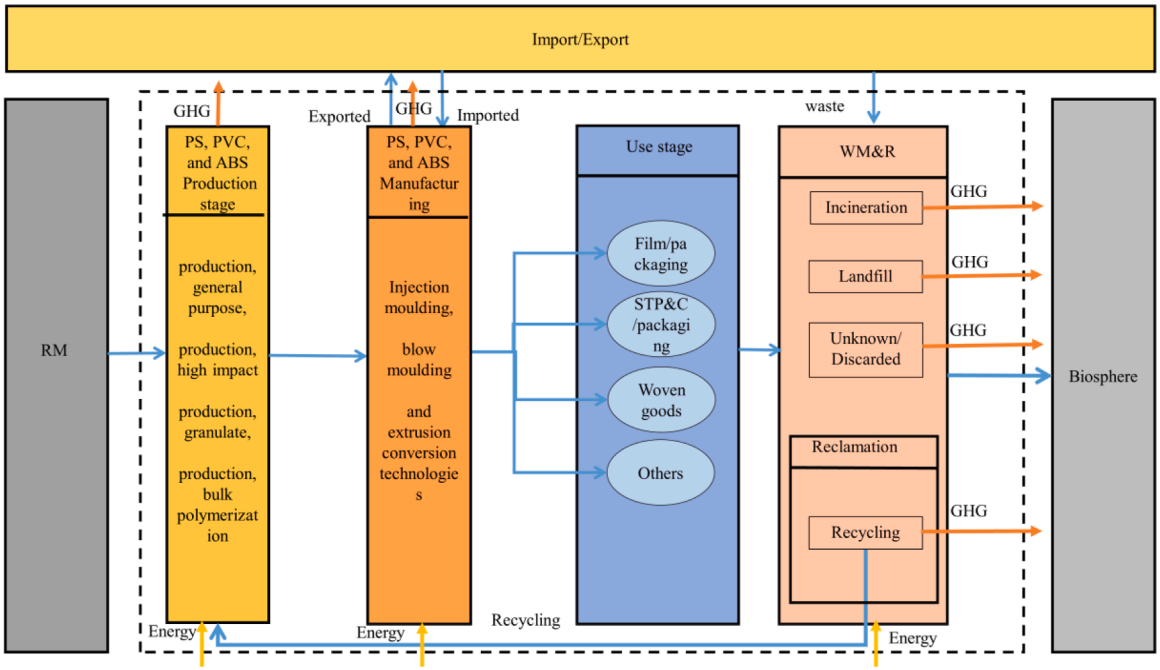


Fig. 1. The system boundaries of material flows and life cycle processes of PS, PVC, and ABS plastics. Note: RM: raw materials; STP&C (sheets, tubes, profiles, and components)

2013; Wang et al., 2021). We calculate the plastic flows of each product based on the top-down method. The detailed calculation formulas of flows in the P process are given below. Detailed formulas for other stages are provided in supplementary materials.

$$CO_y^p = PR_y^p + IM_y^p - EX_y^p \tag{1}$$

$$CO_y^f = \sum_p^3 \left(CO_y^p \times ar_{f,y}^p \times \left(1 - cr_{f,y}^p\right)\right) \tag{2}$$

CO_y^p, PR_y^p, IM_y^p, and EX_y^p represent the apparent consumption, domestic production of primary plastic, the import of primary plastic, and the export of primary plastic p in the year y. CO_y^f refers to the apparent consumption of product f in the year y; $cr_{f,y}^p$ is the scrap rates in the manufacturing stage; $ar_{f,y}^p$ is the product split ratio, and it refers to the percentage of plastic p flowing into product f.

The following formula was used to obtain the GHG emissions of PS, PVC, and ABS plastics in a specific year:

$$GHG_r^y = \sum_1^k E_{r,i,k}^y \times Q^y{}_r \tag{3}$$

GHG_r^y represents GHG emissions during the year y, $E_{r,i,k}^y$ indicates the amount of GHG emitted by PS, PVC, and ABS plastics per unit mass in the year y during the k stage of plastics (quantified using LCA); $Q^y{}_r$ represents the annual production or amount of waste of PS, PVC, and ABS plastics in China (quantified using MFA) (Wang et al., 2021); i represents different GHG, such as carbon dioxide, nitrous oxide, and methane; r represents petroleum-based PS, PVC, and ABS plastics; y represents the year from 2007 to 2017; k represents the number of plastic life cycle stages, and from 1 to 3 represents the P stage, PM stage, WM&R stage respectively.

2.3. Data sources

The data involved in this study are the relevant data required for MFA of PS, PVC, and ABS plastics (such as plastic consumption, consumption structure, plastic waste treatment proportion, etc.), and these data would prepare for the data of accounting activity level. And the emission factor data required for accounting for GHG emissions is needed. By consulting statistical yearbooks, customs statistical information, relevant literature, and industry reports, we can obtain the yield, import, and export volume of PS, PVC, and ABS plastics in the P stage. The proportion and consumption of PS, PVC, and ABS plastic products (such as film, wood goods, sheets, tubes, profiles & components (STP&C), foam plastics, etc.) in the PM stage and the proportion and amount of plastic waste incinerated, landfilled, recycled, untreated/discarded in the WM&R stage are obtained (Chu et al., 2021, CNTIF, 2008-2018; CPCIA, 2008-2018; CPPIA, 2008-2018; Jiang et al., 2020; SFI, 2019). Emission factor data are mainly from the Chinese life cycle database (CLCD), ecoinvent 3.4 database, literature, and European life cycle database (CLCD; European Commission & JRC, 2018; Wernet et al., 2016; Zheng and Suh, 2019). Due to the lack of some emission factor data for PS, PVC, and ABS plastics in China, we can only estimate based on European database. Due to the unavailability of data, uncertainty is inevitable in this study. This study analyzed the effect of uncertainty on results in the supplementary materials. See Table S1-S7, S12 for detailed data sources and uncertainty analysis.

2.4. Scenario analysis of carbon peak in 2030

In October 2021, China issued a guideline to achieve carbon peak and neutrality, requiring that by 2025, an economic system of green and low-carbon circular development would be initially completed, and the energy utilization efficiency of key industries would be greatly improved. The CO_2 emission per unit of GDP would be 18% in 2025 lower than that in 2020, and the consumption proportion of non-fossil energy would reach about 20% in 2025, laying a solid foundation for achieving carbon peak and carbon neutralization (The State Council, 2021). Based on this opinion, this study sets up a scenario analysis of energy structure and plastics consumption transformation, focusing on the carbon peak of the PS, PVC, and ABS plastics in the life cycle in 2030. The first is the baseline scenario, which represents the development trend of energy structure and plastic consumption structure without any human policy interference according to the current trend of development. There are three scenarios of energy structure transformation and

plastic consumption change. In particular, given the guidance's strong mandate, changes in the energy mix need to be considered in all three scenarios. That is to explore the impact of the change of plastic consumption on the realization of the carbon peak in the case that the energy structure target is bound to be achieved. Scenario 1 is that the proportion of non-fossil energy consumption will reach 20% in 2025, the consumption of plastics remains unchanged; scenario 2 shows that the proportion of fossil energy is about 80% in 2025, and the consumption of plastics remains declining; scenario 3 shows that in 2025, the proportion of non-fossil energy is about 20%, and the consumption of plastics remains rising slowly. See Table S8 for the specific scenario explanation.

3. Results and Discussion

3.1. Characteristics of cumulative GHG emissions in the PS, PVC, and ABS life cycle in 2007 - 2017

A material metabolism model was established to show the mass flow of PS, PVC, and ABS from production to disposal, and provide activity level data required for GHG emissions (Fig. 2). Detailed data and calculations were provided in the supplementary materials. Based on the results, the life cycle GHG emissions of these plastics were further analyzed. The study found that the cumulative GHG emissions from PS, PVC, and ABS in the P stage were 163.58, 492.99, and 87.60 Tg CO_2e respectively from 2007 to 2017, accounting for 71.0%, 72.8%, and 61.2% of the total emissions respectively (Fig. 3). The main source of GHG emissions of PS, PVC, and ABS plastics life cycle is the P stage. The GHG emissions from the PM stage are second only to those in the P stage, with 45.37, 123.58, and 37.26 Tg CO_2e respectively, accounting for 19.7%, 16.8%, and 26.0% of the total emissions in the whole life cycle. It is worth noting that the key paths of GHG emissions in the PM stage are different for PS, PVC, and ABS plastics. For example, for PS plastics, the manufacturing processes of PS STP&C and foam plastics lead to a large amount of GHG, and their emissions account for 39.1% and 32.2% of the emissions in the PM stage, reaching 17.76 and 14.63 Tg CO_2e respectively (Fig. 4). Different from PS plastics, the GHG emissions from the manufacturing process of PVC film and STP&C plastics are the main contributions of the PM stage of PVC, with their emissions reaching 19.34 and 45.97 Tg CO_2e respectively, accounting for 15.7% and 37.2% in this stage. It is worth noting that the manufacturing process of PVC used for others also produces a large amount of GHG, with their emissions reaching 33.12 Tg CO_2e. Similar to PS and PVC plastics, the manufacturing process of ABS plastics STP&C contributes the main GHG in this stage. Notably, its emissions account for 84.3% of the emissions in the PM stage. Therefore, the manufacturing processes of PS STP&C, PS foam plastics, PVC film, PVC STP&C, and ABS STP&C are the main contributor to GHG emissions in the PM process, and we suggest that strict reduction policies of GHG emissions should be adopted for the manufacturing process of these products. Nicholson et al. (2021) found that 48 MT CO_2e is produced in the production process during the annual plastic life cycle in the US. In contrast, China produced 121.08 Tg CO_2e in the plastic P stage in 2017. This may be due to the difference in energy structure between China and the US. China has a higher proportion in coal than the US, while the US has a higher proportion in natural gas and nuclear energy (International Energy Agency 2020; Liu et al., 2009).

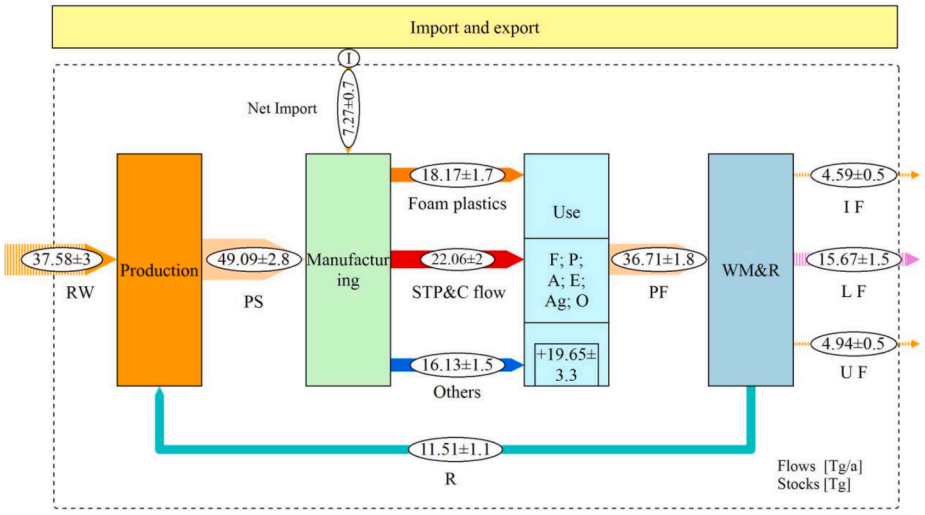
Note: RW: Raw materials; STP&C: sheets, tubes, profiles, and components; PF: post-consumer plastic flow; IF: incineration flow; LF: landfill flow; UF: untreated waste flow; R: recycling; F, P, A, E, Ag, O: building and construction (B&C); packaging, automobiles, electronics, agriculture, and others sectors; WM&R: Waste management and recycling; Others: Synthetic leather, cables, and foam plastics, etc.

Compared with the P and PM stages, the emissions from the WM&R stage are smaller. In 2007-2017, the emissions of GHG in the whole life cycle of PS, PVC, and ABS plastics were 21.39, 60.87, and 18.25 Tg CO_2e respectively. For PS plastics, the cumulative GHG produced by landfill, incineration, recycling processes, and untreated/discarded plastics from 2007 to 2017 were 1.39, 6.08, 10.43, and 3.49 Tg CO_2e respectively. Among them, the incineration and recycling processes of PS plastics produced the most GHG. From 2007 to 2017, the GHG produced by ABS plastic landfill, incineration, recycling processes, and untreated/discarded plastics were 1.18, 5.68, 8.64, and 2.76 Tg CO_2e respectively. Similar to PS plastics, the incineration and recycling processes were the main stage of GHG generation at this stage. The cumulative GHG produced by landfill, incineration, recovery processes, and untreated/discarded plastics of PVC were 3.96, 19.19, 28.89, and 8.82 Tg CO_2e from 2007 to 2017, respectively. It is worth noting that the GHG emissions from the processes of incineration and recovery for PVC plastics are amazing, which are 3.2 and 2.8 times that of PS plastics and 3.4 and 3.3 times that of ABS plastics respectively.

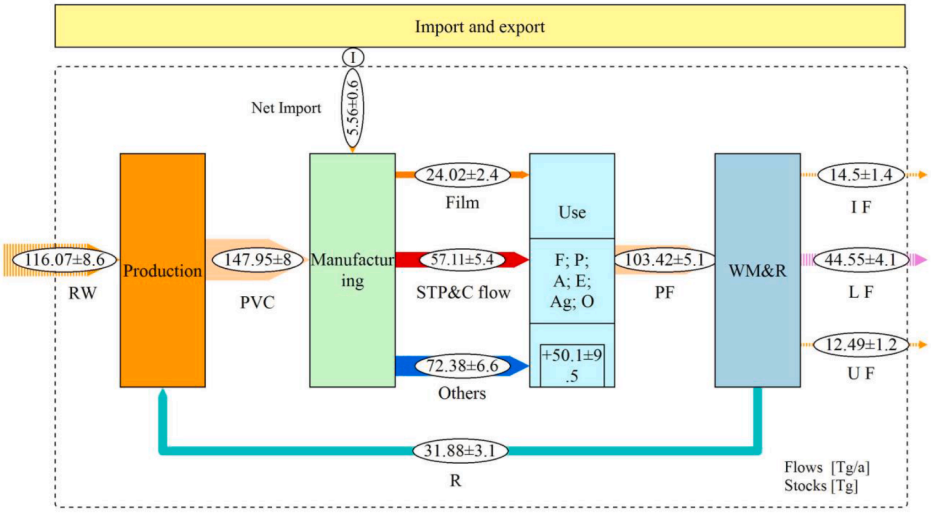
3.2. Characteristics of GHG emissions of PS, PVC, and ABS from 2007 to 2017

The life-cycle GHG emissions of PS, PVC, and ABS in China from 2007 to 2017 are shown in Fig. 5(a). In 2007-2017, the GHG emissions of PS, PVC, and ABS in the whole life cycle were 230.34, 677.43, 143.12 Tg CO_2e respectively. In comparison, the PVC life cycle produced the largest GHG emissions in this period. Life-cycle GHG emissions of PS showed a trend of slowly increasing at first and then slowly decreasing (except 2009) from 2007 to 2017. From 2007 to 2014, the GHG emissions of PS increased slowly with an average growth rate of 8.8%, reaching the maximum value (26.93 Tg CO_2e) in 2014. However, it decreased with an average reduction rate of 9.7% after 2014, and the emissions reached 15.94 Tg CO_2e by 2017. In terms of life cycle stages, the GHG emissions from the PS P stage showed a trend of the first decline, then rise, and then decline from 2007 to 2017. From 2007 to 2009, GHG emissions in the P stage decreased from 11.06 to 9.70 Tg CO_2e. This should be due to the decline in the yield of PS plastics due to the impact of the financial crisis. GHG emissions in the PS P stage showed a slow rise from 2009 to 2016. It is worth noting that its growth rate gradually decreased from 28.8% in 2009 to 1.9% in 2016. This shows that although the GHG emissions in the P stage are rising, their growth is becoming slower and slower. Until 2017, it showed a downward trend, which may be affected by the ban on limiting plastic production. This shows that one of the potentials for carbon neutralization in the PS P stage is to reduce unnecessary production in this stage. For the PS PM stage, the GHG emissions in the STP&C PM stage did not fluctuate significantly from 2007 to 2017, basically remained within the range of 1.1-2.0 Tg CO_2e, and decreased significantly in 2017, which was also affected by plastic limit order (Fig. 5). Similar to PS STP&C, the GHG emissions of PS foam plastics and other products in the PM stage did not fluctuate greatly from 2007 to 2017. However, it decreased significantly in 2009 and 2017. In terms of the time series for the WM&R stage, GHG emissions from landfills increased at an average rate of 20.9% from 2007 to 2017 (except for 2009 and 2017), with a particularly large increase from 2009 to 2010 (Fig. 6). Different from the landfill process, GHG emissions from the incineration process showed an upward trend (except 2009), with an average increase rate of 47.3% from 2007 to 2017. Despite the impact of the plastic restriction order in 2017, which reduced the total amount of waste, the amount of waste incinerated increased, so the emissions from 2016 to 2017 were still rising. From 2007 to 2017, the emissions from the recycling process showed a rapid increase after 2009 and then a slow increase. The possible GHG brought by untreated plastics wastes showed a gradual downward trend from 2007 to 2017, which should be that with the improvement of the waste management system, untreated plastic wastes were gradually incinerated or landfilled.

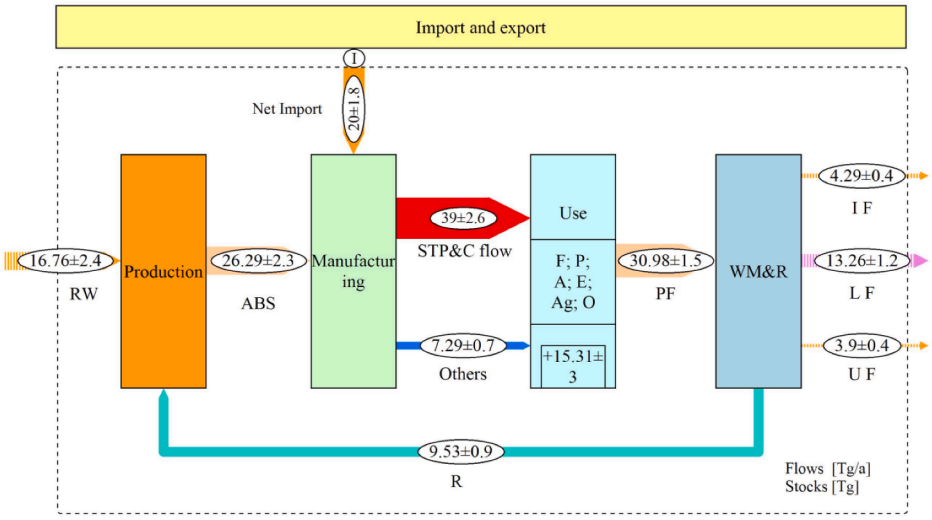
Different from PS, the life-cycle GHG emissions of PVC from 2007 to 2017 showed a trend of slow decline at first and then slow rise. From 2007 to 2008, the GHG emission of PVC in the whole life cycle decreased from 43.77 Tg CO_2e to 39.69 Tg CO_2e, but after 2008, it increased at an



(a) MFA mass flow of PS in the life cycle in 2007-2017



(b) MFA mass flow of PVC in the life cycle in 2007-2017



(c) MFA mass flow of ABS in the life cycle in 2007-2017

Fig. 2. MFA mass flows of PS, PVC, and ABS life cycle in 2007-2017 in China.

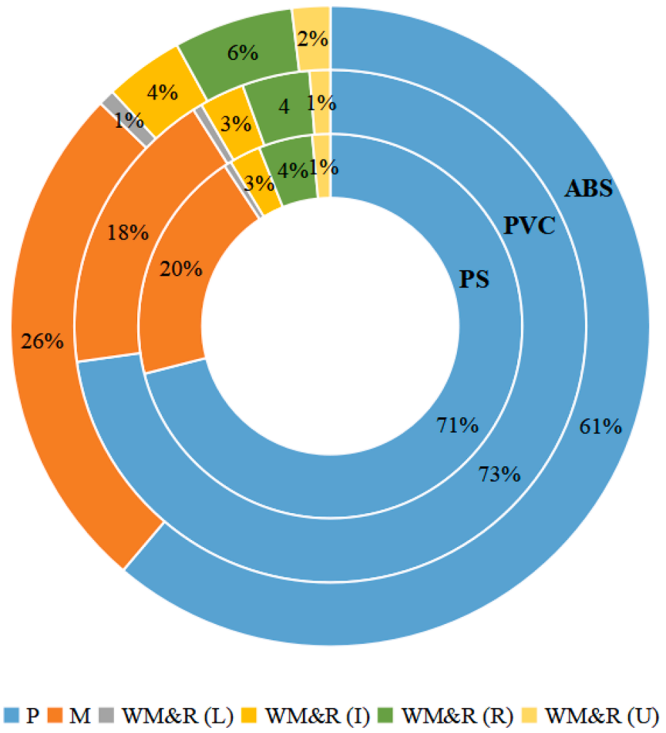


Fig. 3. Proportion of GHG emissions at all stages of PS, PVC, and ABS plastics in 2007-2017

average increase rate of 9.2%, and by 2017, the emission reached 85.42 Tg CO_2e. The emission characteristics of each stage of PVC are different from those of PS from 2007 to 2017. In addition to 2008, the GHG emissions in the PVC P stage showed an upward trend at an average growth rate of 8.4% after 2009, and the GHG emissions also increased from 30.52 Tg CO_2e in 2009 to 59.64 Tg CO_2e in 2017. In the PM stage, the GHG emissions from the manufacturing of PVC film, synthetic leather, STP&C, cables, and other products during this period reached lowest point in 2008 and then showed a large increase until 2013. However, since 2014, although it was also increasing, the growth rate was not large. These characteristics are inconsistent with the PS plastics PM stage. In the stage of WM&R, the GHG emissions from the processes of the landfill, incineration, recycling, and untreated/discarded plastics were underestimated in 2008 and increased rapidly in 2010 and 2017. And it reached 0.73, 6.49, 4.03, and 0.20 Tg CO_2e respectively by 2017. Different from landfill, incineration and recycling, GHG emissions from untreated plastic wastes decreased in 2010, 2015, and 2017.

The trend of life-cycle GHG emission of ABS is very different from that of PS and PVC. From 2007 to 2017, the GHG emissions of ABS kept a steady upward trend, from 8.71 Tg CO_2e in 2007 to 19.72 Tg CO_2e in 2017, with an average annual growth rate of 8.7%. The emission characteristics of the ABS life cycle are different from PS and PVC from 2007 to 2017. In the P stage, GHG emissions showed an upward trend with an average growth rate of 9.9%, and the amount of GHG also increased from 4.76 Tg CO_2e in 2007 to 11.97 Tg CO_2e in 2017. During the PM stage, the GHG emissions from the manufacturing of ABS STP&C and other products increased rapidly in 2014. In the WM&R stage, the GHG emissions in the processes of the landfill, incineration, recycling, and untreated in this period reached a low point in 2012 as a result of policy interventions.

3.3. Scenario analysis of carbon peak in 2030

Under the baseline scenario, the GHG emissions of PS, PVC, and ABS plastics does not reach the carbon peak in 2030 and may reach 23.17, 146.41, and 42.31 Tg CO_2e respectively in 2030 (Fig. 7), which will be 1.5, 1.7, and 2.1 times that in 2017. Under the scenario of changing the energy structure and plastics production, in scenario 1, the GHG emissions of PS will achieve a carbon peak before 2030 and achieve a carbon emission reduction of 3.45 Tg CO_2e in 2035 compared with 2017. Compared with the baseline scenario, the GHG emissions of scenario 1 in 2030 and 2035 are reduced by 39% and 53% respectively. Scenario 2 has greater potential for carbon emission reduction. Since 2030, GHG emissions show a rapid downward trend (Fig. 8). By 2035, GHG emissions will be reduced by 8.47 Tg CO_2e compared with 2017. Moreover, GHG emissions in 2030 and 2035 are reduced by 38% and 72% respectively compared with the baseline scenario. In scenario 3, carbon peak can also be achieved before 2030, and GHG emissions will remain constant after 2030. Compared with the baseline scenario, GHG emissions in scenario 3 will be reduced by 38% and 47% in 2030 and 2035, respectively. Similar to PS plastics, under the scenario of changing the energy structure and plastics yield, in scenario 1, the GHG emissions for PVC will achieve a carbon peak before 2030, and a carbon emission reduction of 6.39 Tg CO_2e in 2035 compared with 2017. Moreover, GHG emissions in 2035 under scenario 1 are reduced by 56% compared with the baseline scenario. The carbon emission reduction potential in scenario 2 is the most significant (Fig. 8). By 2035, the GHG emissions will be reduced by 38.24 Tg CO_2e compared with that in 2017, which is 6 times that in scenario 1. It is worth noting that GHG emissions in 2030 and 2035 under scenario 2 are reduced by 38% and 74% respectively compared with the baseline scenario. The carbon peak also been achieved in scenario 3, but the emission reduction is not as significant as that in scenarios 1 and 2. Under scenario 3, GHG emissions in 2035 are reduced by 50% compared with the baseline scenario. Similarly, the GHG emissions of ABS will achieve the carbon peak goal before 2030, and the emission at the carbon peak is about 27.35 Tg CO_2e in scenario 1. Scenario 2 has the most significant carbon emission reduction potential (Fig. 8). Similar to PS and PVC, ABS plastics in scenario 3 also achieve a carbon peak. Moreover, compared with the baseline scenario, GHG emissions in 2035 under scenarios 1, 2 and 3 are reduced by 61%, 76% and 48% respectively. The analysis shows that the goal of carbon neutralization is also most likely to be achieved under the change of

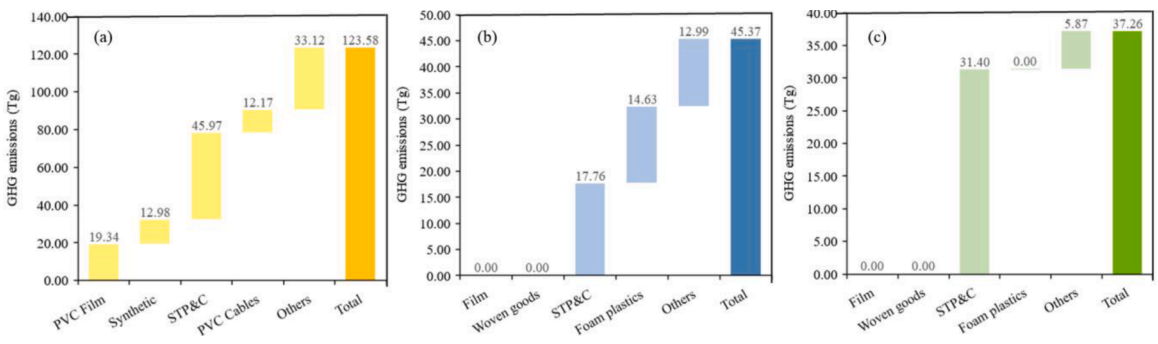


Fig. 4. GHG emissions from products manufacturing in PS, PVC, and ABS plastics PM stage from 2007 to 2017 ((a) PVC, (b) PS, (c) ABS)

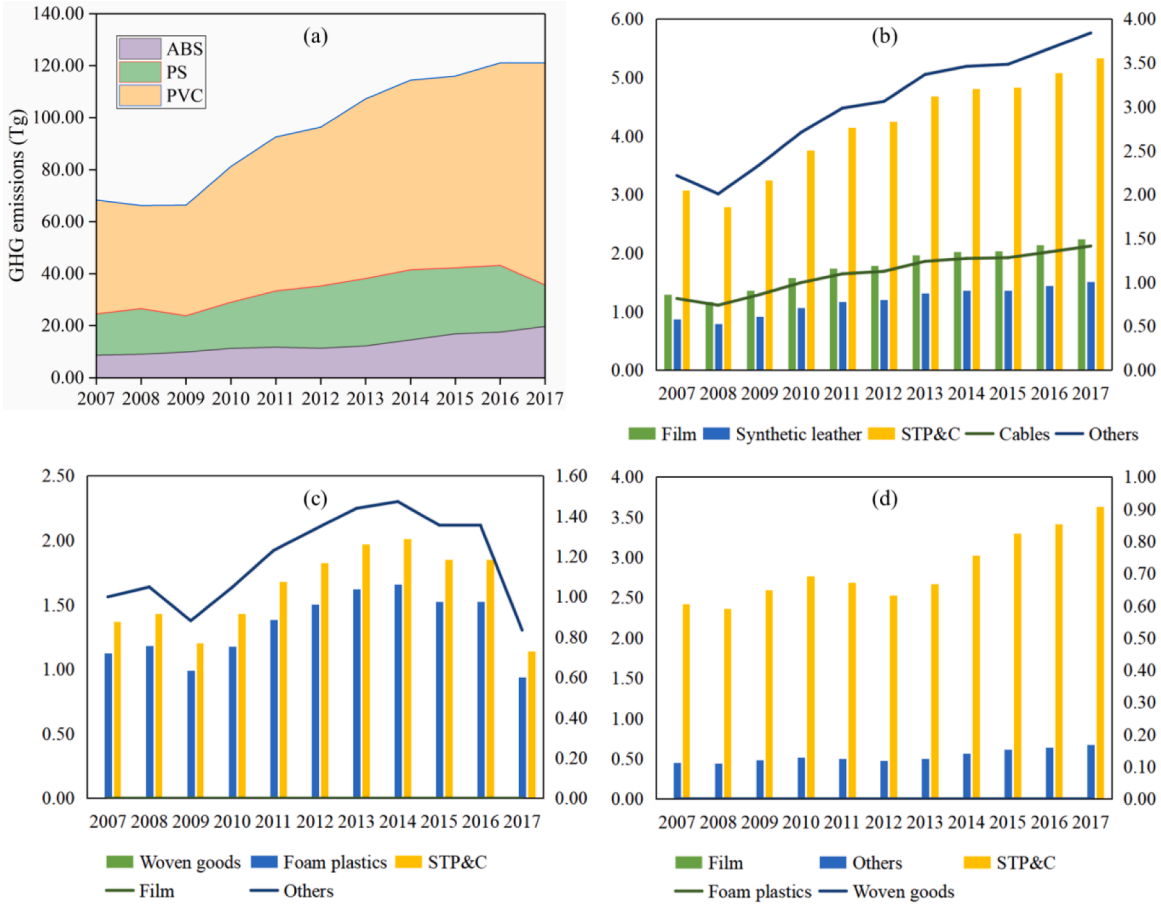


Fig. 5. Total GHG emissions of PS, PVC, and ABS plastics in the whole life cycle and the PM stage from 2007 to 2017 ((a) total GHG emissions, (b) PS, (c) PVC, (d) ABS)

energy structure and plastic yield.

4. Discussion

The P stage is the main source of GHG in the whole life cycle of PS, PVC, and ABS plastics. The PM stage is second only to the P stage, and relatively speaking, the emissions in the WM&R stage are the lowest. Similarly, the production of raw materials (68.4%) is the main stage of environmental impact in the life cycle of express packaging materials, followed by the manufacturing process (18.8%) and the end of life stage (12.7%) (Su et al., 2020). Zheng and Suh, (2019) found that most of the GHG emissions were generated in the plastic production stage, accounting for 61% of the total emissions, followed by the PM stage, accounting for 30%, accounting for 91% in total. This is similar to the results of this study. The P stage and PM stage are also the main emission processes in this study, accounting for 89%. They also found that the global petroleum-based plastics (including 10 kinds) emitted 1.8 $GtCO_2e$ over their life cycle in 2015. We found that China's petroleum-based PS, PVC, and ABS plastics emitted 121.08 $TgCO_2e$ over their life cycle in 2017. Therefore, only these three plastics in China emitted 7% of the global GHG. Nowadays, the demand for plastics is large, and the production of a large number of plastics requires a lot of energy input. The production of polymers consumes a lot of fossil fuel energy during crude oil extraction and refining. GHG sources include direct emissions (such as methane leakage and combustion) from fuel combustion and energy consumption during oil and gas drilling. The manufacturing process of plastic products is energy-intensive and emission-intensive. By cracking alkanes into olefins, polymerizing and plasticizing olefins into plastic resins, and other chemical refining processes, A lot of GHG are generated (Cai et al., 2019;Hamilton et al., 2019). This study also found that, in contrast, the PVC life cycle generated the largest GHG emissions from 2007 to 2017. This is similar to the results of Alsabri and Al-Ghamdi (2020), who show that PVC has higher energy consumption and carbon emission than other plastics.

To reduce GHG emissions, it seems feasible to replace petroleum plastics with bioplastics (Posen et al., 2016;. Semba et al. 2018) found that 20% of sugarcane and 80% of corn produce biological p-xylene, and the GHG emissions of 100% biological derived PET are 24.0% lower than that of petroleum PET. It is worth noting that when considering the landfill and compost emissions of PLA in the waste management stage, its life cycle emissions increase significantly (Benavides et al., 2020). By studying the GHG emissions characteristics of PS, PVC, and ABS plastics on a time scale, it can be found that the GHG emissions increased significantly with the growth of the production of these plastics from 2007 to 2017. Reducing the yield and use of PS, PVC, and ABS plastics is a strong strategy to deal with the climate impact of the whole life cycle of plastics. We recommend some measures to reduce or prohibit the use of non-essential PS, PVC, and ABS plastics products, such as restricting or prohibiting disposable PS, PVC, and ABS plastics (such as PS foam plastics, PVC film, and disposable plastics commonly used in packaging services). At present, China has promulgated the plastics restriction order. Adhering to the plastics restriction order policy will play an important role in promoting the carbon emission reduction of the plastic industry. Stopping the production of petrochemical plastics and the exploitation of fossil fuels are the most effective ways to mitigate the impact on the climate. These solutions need the urgent support of policymakers and charitable funders (Hamilton et al., 2019; Shen et al., 2020). Notably, our research found that the PM sectors of PS STP&C, PS

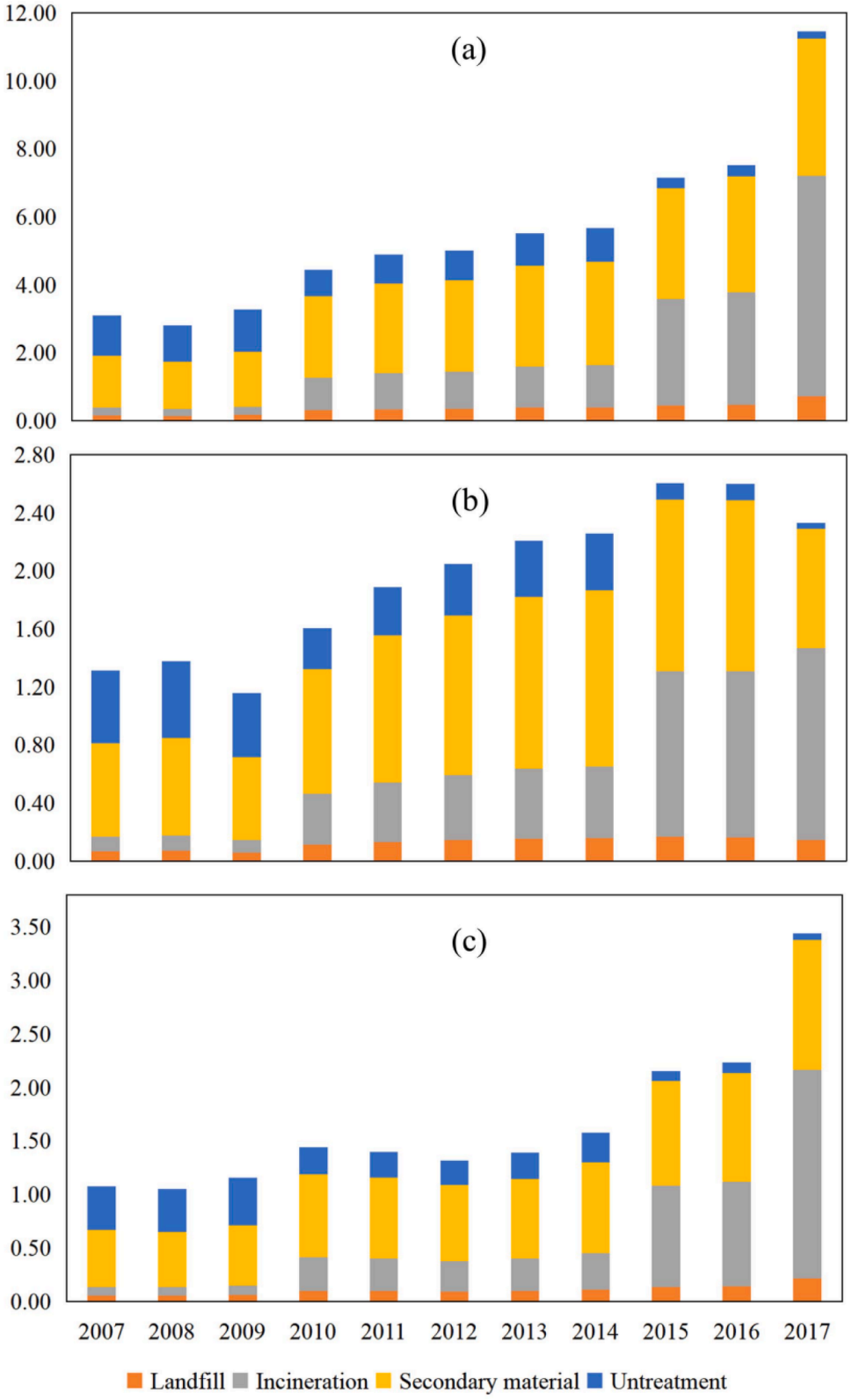


Fig. 6. GHG emissions from PS, PVC, and ABS plastic in the WM&R phases from 2007 to 2017 ((a) PS, (b) PVC, (c) ABS)

foam plastics, PVC film, PVC STP&C, and ABS STP&C are the key sectors with high GHG emissions. Therefore, we need to take strict control measures for these key sectors, focusing on reducing the production of non-essential products in these plastic products and replacing them with bioplastics. We suggest investment in low-carbon energy and high-efficiency technologies in the PM process of these products, and elimination of technologies and processes with low efficiency, high energy consumption, and high emissions. In the WM&R stage, incineration is the main contribution of GHG. Shen et al. (2020) also found that the incineration of waste plastics will become the main source of GHG emissions. Research showed that each tonne of plastic packaging waste would release 790 kg or about 2.9 tonnes of CO_2 into the atmosphere

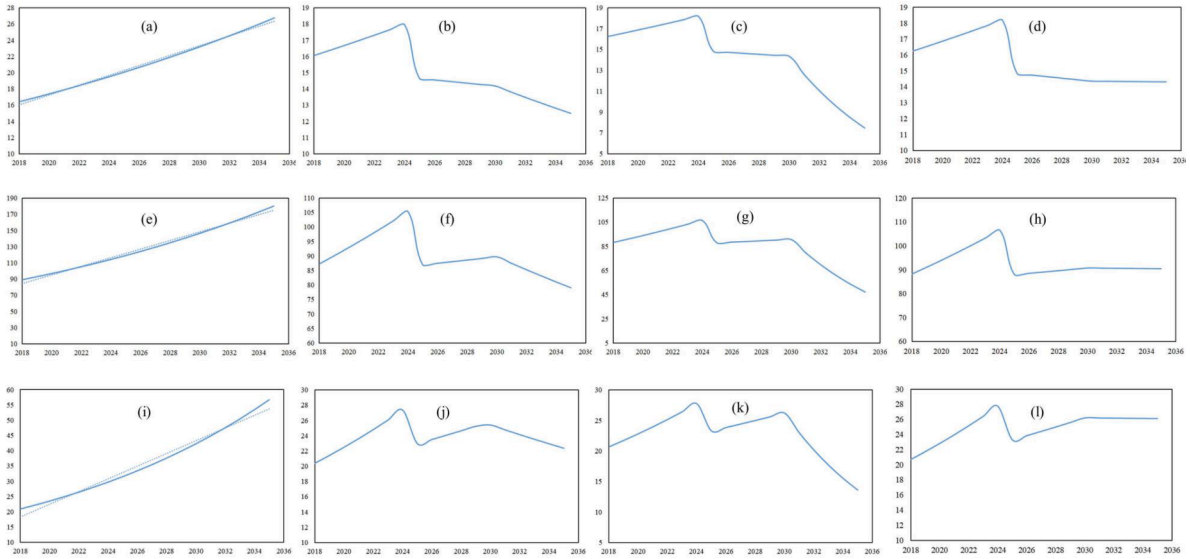


Fig. 7. Characteristics of carbon peak in 2030 under various scenarios (a, b, c, and d represent the baseline scenario, scenarios 1, 2, and 3 of PS respectively; e, f, g, and h represent the baseline scenario, scenarios 1, 2, and 3 of PVC respectively; i, j, k, and l represent the baseline scenario, scenarios 1, 2 and 3 of ABS respectively)

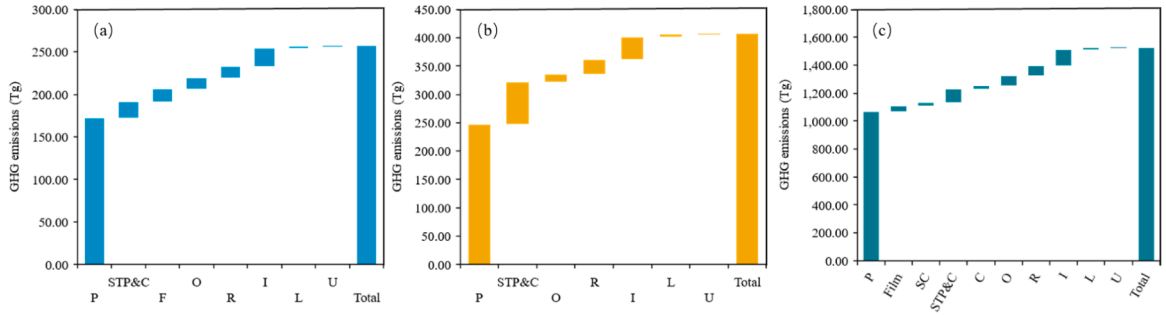


Fig. 8. Life cycle GHG total emissions of PS, PVC, ABS plastics under scenario 2 in 2018 - 2035 (Tg) (a, b, and c represent PS, ABS, and PVC plastics respectively)

(Hamilton et al., 2019). Hamilton et al. (2019) reported that the net GHG emissions from plastic waste incineration in 2015 were about 16 million tonnes. In 2015, the CO_2 emission from waste incineration in the US was 11 million tonnes, and the GHG generated by plastic waste incineration was equivalent to 1.26 million vehicles driving for a year or consuming more than 5 billion gallons of gasoline (US EPA, 2018; Shen et al., 2020). This study showed that in 2017, the GHG generated by the incineration of PS, PVC, and ABS plastic waste would be 9.76 Tg, accounting for 61% and 89% of the above values respectively. Secondly, the recycling process will also produce a large amount of GHG. This is not only a large amount of recovery of PS, PVC, and ABS plastics but also a large amount of fossil energy that needs to be invested in the process of recovery. It is found that since the raw materials of plastic products come from fossil fuels, recycling and reusing plastic wastes will not only reduce the input of fossil fuels but also reduce the related GHG emissions (Liu et al., 2021). Compared with the original plastic products, 130,000 kJ of energy can be saved per kilogram of recycled plastics (Rahimi and García, 2017). Another study shows that recycled plastic materials can replace raw materials and reduce the impact of materials on the environment by 30% (Galve et al., 2021). In general, compared with incineration, recycling is more conducive to emission reduction, but the GHG emissions in the recycling process cannot be ignored. The recovered 3.17 million tonnes of plastic waste can save about 3.2 million tonnes of CO_2e (US EPA, 2016). However, the recycling work still needs to solve some challenges (i.e., low-grade plastic waste similar to multi-layer plastic packaging is difficult to separate and dispose, and colorants, fillers, and additives used in the plastic production process affect recycling, and the recycling cost is high), to obtain the GHG benefits brought by recycling in the long run. In addition, biomass utilization, and carbon capture and utilization technologies have significant effects on reducing GHG emissions during the life cycle of plastics. Meys et al. (2021) found that by combining biomass utilization, recycling, and carbon capture and utilization, compared with the current fossil-based production technology, it can not only achieve net-zero GHG emissions of plastics but reduce energy demand and operation cost.

We found that if according to the current development trend, the GHG emissions from the life cycle of PS, PVC, and ABS plastics will not reach the carbon peak goal in 2030. However, under the scenario of changing the energy structure and plastics production, the GHG in the life cycle of PS, PVC, and ABS plastics has reached the carbon peak before 2030. Zheng and Suh, (2019) found that GHG emissions reduction in the plastic life cycle showed great potential under the scenario of energy decarbonization. By 2050, switching to renewable energy sources will reduce GHG emissions by an average of 62%. Our research showed that by 2030, the optimal scenario of changing energy structure and production will reduce GHG emissions by 38% compared with the baseline scenario. Therefore, we must reduce the use of fossil energy, increase the consumption proportion of non-fossil energy (at least 20% in 2025), and limit the consumption of unnecessary plastics (the growth rate of plastics will remain unchanged or decrease by 10% year by year after 2030), which has a very significant benefit for the plastic industry to achieve carbon peak. The lack of this study is mainly due to the unavailability of data, the GHG emissions of PS, PVC, and ABS plastics after 2017 have not been analyzed, and the GHG emissions in the use stage of

plastics have not been considered. By improving the database, exploring the GHG emissions in the use stage of plastics after 2017 will be the future research direction. In addition, the GHG emissions generated by discarded waste and plastics in the use phase are also one of the future research directions.

5. Conclusions

This study combines dynamic MFA with LCA to establish a time series analysis model for life-cycle GHG emissions from PS, PVC, and ABS plastics in China. Further, the paths and strategies of carbon peak in 2030 are analyzed. The main conclusions are as follows:

- The main sources of life-cycle GHG for PS, PVC, and ABS are the P and PM stages. The key paths of GHG emissions are the PM processes of PS STP&C, PS foam plastics, PVC film, PVC STP&C, and ABS STP&C. Therefore, the manufacturing of these plastics requires stricter policies (e.g., limiting disposable PS foam and PVC film) to reduce emissions.
- The GHG emissions from incineration and recycling cannot be ignored. The cumulative GHG emissions of PS, PVC, and ABS reached 230.34, 677.43, 143.12 Tg CO_2e in 2007-2017, respectively. GHG emissions of PS showed a trend of slow rise and then slow decline from 2007 to 2017. Unlike PS, the emissions of ABS maintained a steady upward trend.
- Under the baseline scenario, the life-cycle GHG of PS, PVC, and ABS could not reach the carbon peak in 2030. Increasing the proportion of non-fossil energy (at least 20% in 2025) and limiting the growth rate of plastics (it remains unchanged or decrease by 10%) will help to reach peak carbon before 2030, and the optimal scenario will reduce GHG emissions by 38% in 2030.
- According to the results, this study identifies the major sources of life-cycle GHG emissions from PS, PVC, and ABS plastics, and reveals the paths and strategies for carbon peak by 2030.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that cdocould have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.resconrec.2022.106295.

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